Tugas Praktikum Pemrograman Mobile
Pertemuan Ke - 4



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Praktikum Flutter

1. Tujuan

Praktikum 4 ini bertujuan agar kita memahami proses pengambilan data secara asynchronous menggunakan Future, async, dan await, mengambil data dari REST API menggunakan package http, mengubah JSON menjadi object Dart melalui fromJson, menampilkan data berdasarkan status loading, error, dan data menggunakan FutureBuilder, serta menangani kegagalan jaringan dengan tombol.

2. Alat dan Bahan

- Komputer Atau Laptop
- Flutter Package (Flutter - Install)
- Android Studio / IDE Dengan Support Flutter & Dart
- Emulator SDK Android / HP Android Dengan Debugging Mode
- Koneksi Internet Untuk Mengakses JSONPlaceHolder API

3. Percobaan Koneksi REST API

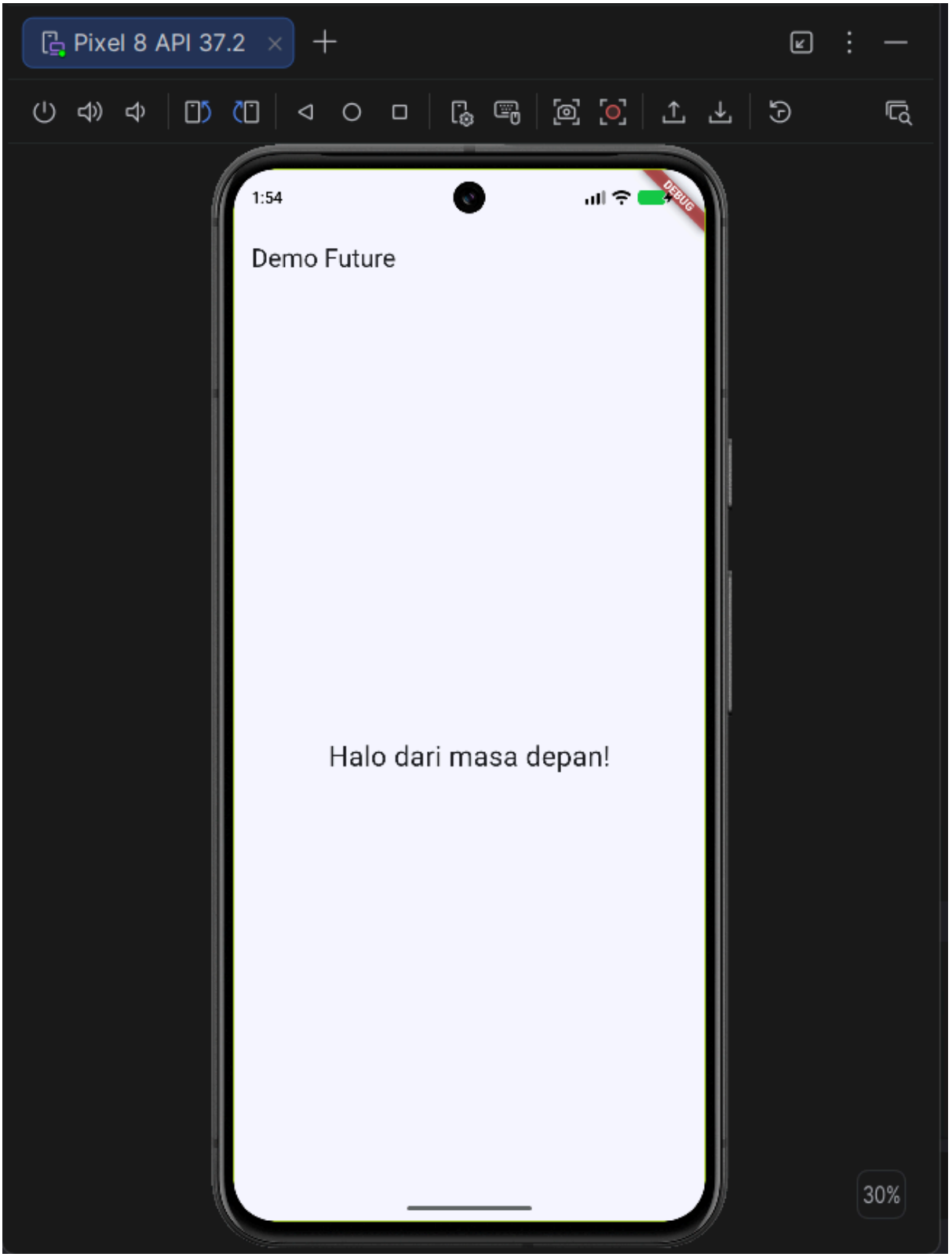
3.1 Mengenal Future dan FutureBuilder :

main.dart

```dart
import 'package:flutter/material.dart';

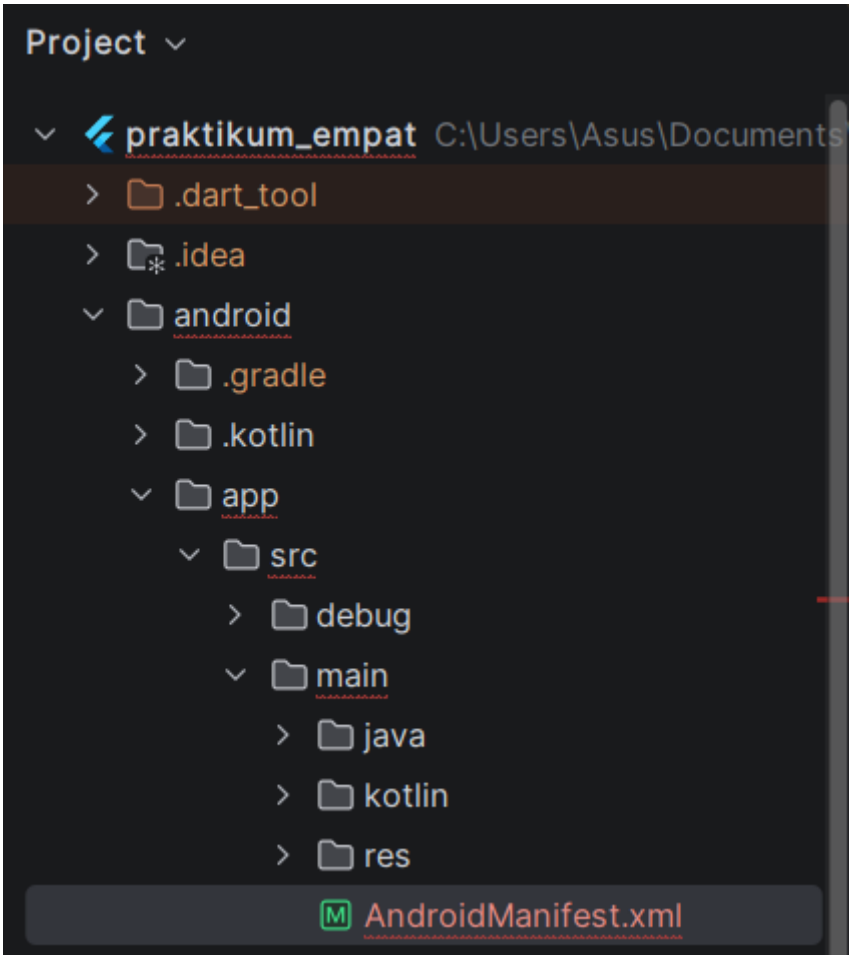
void main() => runApp(const MyApp());
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 4',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const SalamPage(),
    ); MaterialApp
  }
}
Future<String> ambilSalam() async {
  await Future.delayed(const Duration(seconds: 2));
  return 'Halo dari masa depan!';
}
```

```dart
class SalamPage extends StatefulWidget {
  const SalamPage({super.key});
  @override
  State<SalamPage> createState() => _SalamPageState();
}
class _SalamPageState extends State<SalamPage> {
  late Future<String> _future;
  @override
  void initState() {
    super.initState();
    _future = ambilSalam();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Demo Future')),
      body: Center(
        child: FutureBuilder<String>(
          future: _future,
          builder: (context, snapshot) {
            if (snapshot.connectionState == ConnectionState.waiting) {
              return const CircularProgressIndicator();
            }
            if (snapshot.hasError) {
              return Text('Error: ${snapshot.error}');
            }
            return Text(snapshot.data!, style: const TextStyle(fontSize: 24));
          },
        ), FutureBuilder
      ), Center
    ); Scaffold
  }
}
```


Pixel 8 API 37.2
1:54
DEBUG
Demo Future
Halo dari masa depan!
30%

3.2 Menyiapkan Paket HTTP :

```
PS C:\Users\Asus\Documents\University\Semester 5\Pemrograman Mobile\Praktikum\praktikum_empat> flutter pub add http
Resolving dependencies...
Downloading packages...
  cupertino_icons 1.0.9 (2.0.0 available)
+ http 1.6.0
+ http_parser 4.1.2
  material_color_utilities 0.13.0 (0.13.1 available)
  meta 1.18.3 (1.19.0 available)
  test_api 0.7.12 (0.7.14 available)
+ typed_data 1.4.0
  vector_math 2.4.0 (2.4.3 available)
+ web 1.1.1
Changed 4 dependencies!
```



AndroidManifest.xml

```
1  <manifest xmlns:android="http://schemas.android.com/apk/res/android">
2      <uses-permission android:name="android.permission.INTERNET" />
3      <application
4          android:label="praktikum_empat"
5          android:name="${applicationName}"
6          android:icon="@mipmap/ic_launcher">
```

3.3 Penambahan Model dan Fungsi Pengambilan Data :

main.dart

```dart
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

class Pengguna {
  final int id;
  final String name;
  final String email;
  final String phone;
  final String website;

  const Pengguna({
    required this.id,
    required this.name,
    required this.email,
    required this.phone,
    required this.website,
  });
  factory Pengguna.fromJson(Map<String, dynamic> json) {
    return Pengguna(
      id: json['id'] as int,
      name: json['name'] as String,
      email: json['email'] as String,
      phone: json['phone'] as String,
      website: json['website'] as String,
    );
  }
}
Future<List<Pengguna>> ambilPengguna() async {
  final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
  final response = await http.get(uri).timeout(const Duration(seconds: 10));
  if (response.statusCode != 200) {
    throw Exception('Gagal memuat data (kode ${response.statusCode})');
  }
  final List<dynamic> data = jsonDecode(response.body);
  return data
      .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
      .toList();
}

void main() => runApp(const MyApp());
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 4',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const SalamPage(),
    );
```

MaterialApp

```dart
  }
}
Future<String> ambilSalam() async {
  await Future.delayed(const Duration(seconds: 2));
  return 'Halo dari masa depan!';
}
class SalamPage extends StatefulWidget {
  const SalamPage({super.key});
  @override
  State<SalamPage> createState() => _SalamPageState();
}
class _SalamPageState extends State<SalamPage> {
  late Future<String> _future;
  @override
  void initState() {
    super.initState();
    _future = ambilSalam();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Demo Future')),
      body: Center(
        child: FutureBuilder<String>(
          future: _future,
          builder: (context, snapshot) {
            if (snapshot.connectionState == ConnectionState.waiting) {
              return const CircularProgressIndicator();
            }
            if (snapshot.hasError) {
              return Text('Error: ${snapshot.error}');
            }
            return Text(snapshot.data!, style: const TextStyle(fontSize: 24));
          },
        ), FutureBuilder
      ), Center
    ); Scaffold
  }
}
```

3.4 Menampilkan Data FutureBuilder dan Halaman Detail :

main.dart

```dart
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

class Pengguna {
  final int id;
  final String name;
  final String email;
  final String phone;
  final String website;

  const Pengguna({
    required this.id,
    required this.name,
    required this.email,
    required this.phone,
    required this.website,
  });
  factory Pengguna.fromJson(Map<String, dynamic> json) {
    return Pengguna(
      id: json['id'] as int,
      name: json['name'] as String,
      email: json['email'] as String,
      phone: json['phone'] as String,
      website: json['website'] as String,
    );
  }
}
Future<List<Pengguna>> ambilPengguna() async {
  final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
  final response = await http.get(uri).timeout(const Duration(seconds: 10));
  if (response.statusCode != 200) {
    throw Exception('Gagal memuat data (kode ${response.statusCode})');
  }
  final List<dynamic> data = jsonDecode(response.body);
  return data
      .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
      .toList();
}

class DetailPenggunaPage extends StatelessWidget {
  final Pengguna pengguna;
  const DetailPenggunaPage({super.key, required this.pengguna});
  @override
```

```dart
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(pengguna.name)),
      body: ListView(
        children: [
          ListTile(
            leading: const Icon(Icons.email),
            title: Text(pengguna.email),
          ), ListTile
          ListTile(
            leading: const Icon(Icons.phone),
            title: Text(pengguna.phone),
          ), ListTile
          ListTile(
            leading: const Icon(Icons.language),
            title: Text(pengguna.website),
          ), ListTile
        ],
      ), ListView
    ); Scaffold
  }
}


class PenggunaPage extends StatefulWidget {
  const PenggunaPage({super.key});

  @override
  State<PenggunaPage> createState() =>
      _PenggunaPageState();
}

class _PenggunaPageState
    extends State<PenggunaPage> {
  late Future<List<Pengguna>> _future;

  @override
  void initState() {
    super.initState();
    _future = ambilPengguna();
  }

  void _muatUlang() {
    setState(() {
```

```
    _future = ambilPengguna();
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Daftar Pengguna'),
      actions: [
        IconButton(
          icon: const Icon(Icons.refresh),
          onPressed: _muatUlang,
        ), IconButton
      ],
    ), AppBar
    body: FutureBuilder<List<Pengguna>>(
      future: _future,
      builder: (context, snapshot) {
        if (snapshot.connectionState ==
            ConnectionState.waiting) {
          return const Center(
            child: CircularProgressIndicator(),
          ); Center
        }

        if (snapshot.hasError) {
          return Center(
            child: Padding(
              padding: const EdgeInsets.all(24),
              child: Column(
                mainAxisSize:
                MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.error_outline,
                    size: 48,
                  ), Icon
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n'
                        '${snapshot.error}',
                    textAlign:
                    TextAlign.center,
                  ), Text
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
```

```
                        child: const Text(
                          'Coba lagi',
                        ), Text
                      ), ElevatedButton
                    ],
                  ), Column
                ), Padding
              ); Center
            }

            final data = snapshot.data!;

            return ListView.builder(
              itemCount: data.length,
              itemBuilder: (context, i) {
                final p = data[i];

                return ListTile(
                  leading: CircleAvatar(
                    child: Text(p.name[0]),
                  ), CircleAvatar
                  title: Text(p.name),
                  subtitle: Text(p.email),
                  trailing: const Icon(
                    Icons.chevron_right,
                  ), Icon
                  onTap: () {
                    Navigator.push(
                      context,
                      MaterialPageRoute(
```

```
                        builder: (_) =>
                            DetailPenggunaPage(pengguna: p),
                      ), MaterialPageRoute
                    );
                  },
                ); ListTile
              },
            ); ListView.builder
          },
        ), FutureBuilder
      ); Scaffold
    }
  }

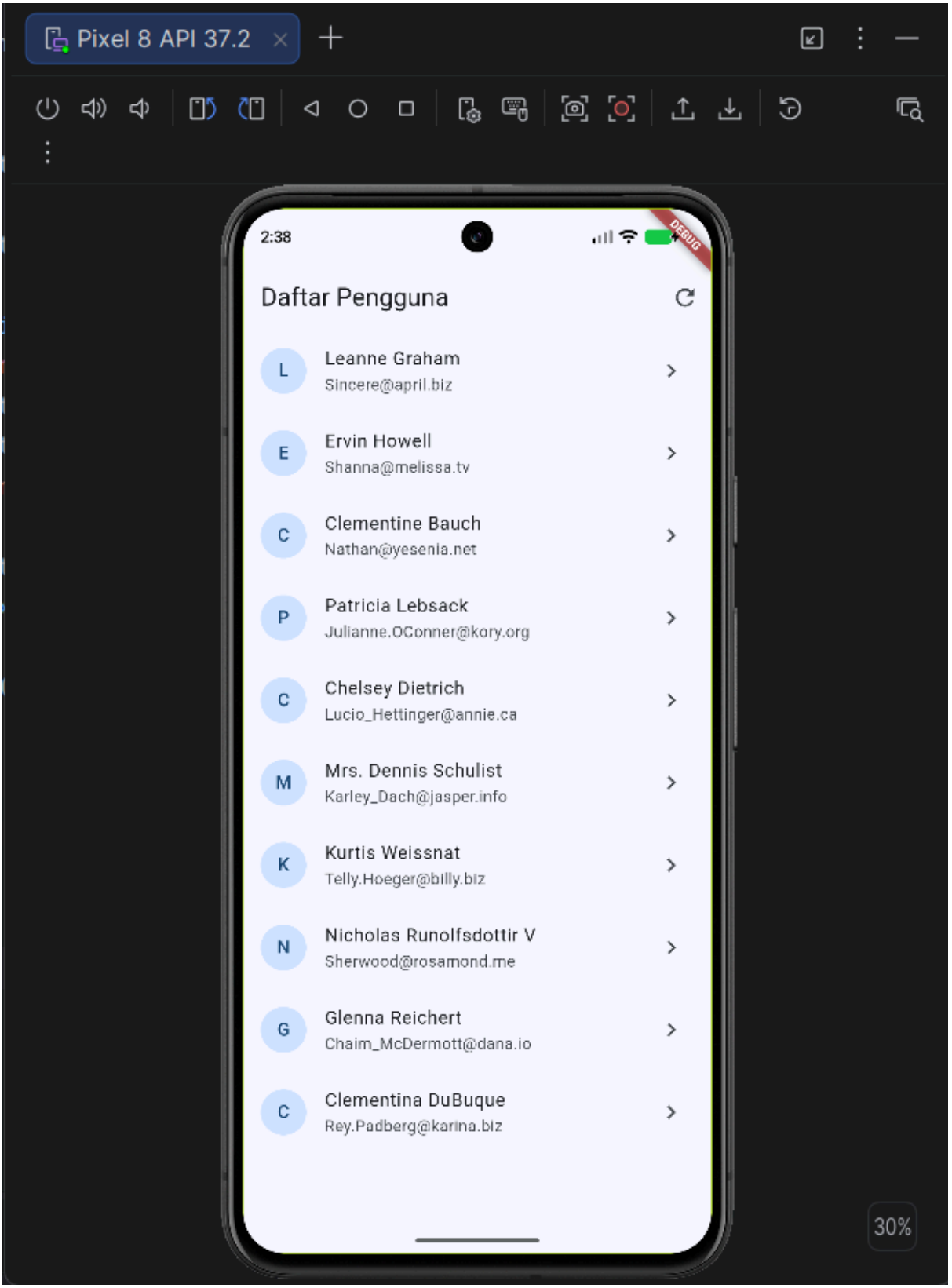
  void main() => runApp(const MyApp());
  class MyApp extends StatelessWidget {
    const MyApp({super.key});
    @override
```

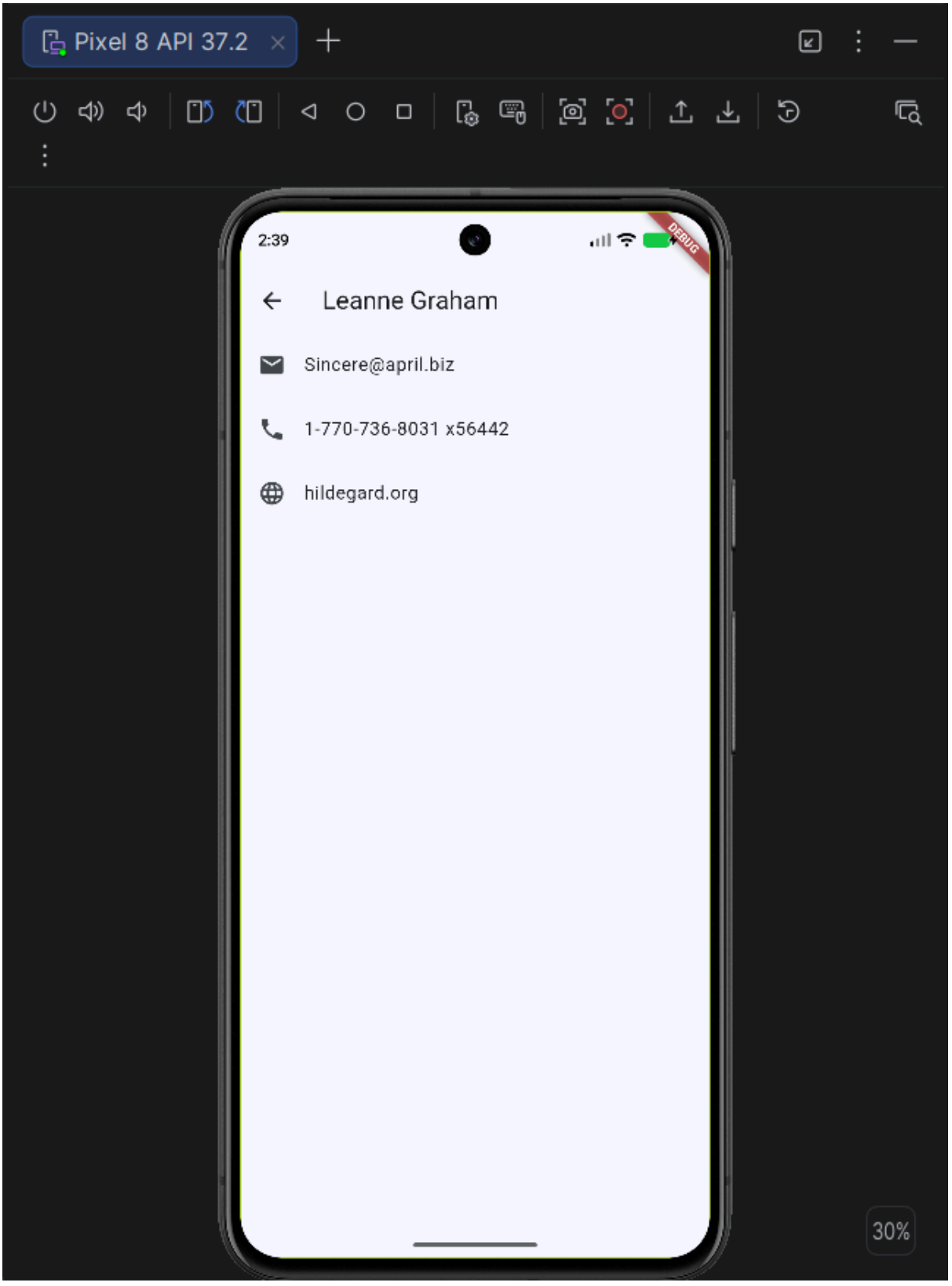
```dart
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 4',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const PenggunaPage(),
    ); MaterialApp
  }
}
Future<String> ambilSalam() async {
  await Future.delayed(const Duration(seconds: 2));
  return 'Halo dari masa depan!';
}
class SalamPage extends StatefulWidget {
  const SalamPage({super.key});
  @override
  State<SalamPage> createState() => _SalamPageState();
}
class _SalamPageState extends State<SalamPage> {
  late Future<String> _future;
  @override
  void initState() {
    super.initState();
    _future = ambilSalam();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Demo Future')),
      body: Center(
        child: FutureBuilder<String>(
```

22

```dart
          future: _future,
          builder: (context, snapshot) {
            if (snapshot.connectionState == ConnectionState.waiting) {
              return const CircularProgressIndicator();
            }
            if (snapshot.hasError) {
              return Text('Error: ${snapshot.error}');
            }
            return Text(snapshot.data!, style: const TextStyle(fontSize: 24));
          },
        ), FutureBuilder
      ), Center
    ); Scaffold
  }
}

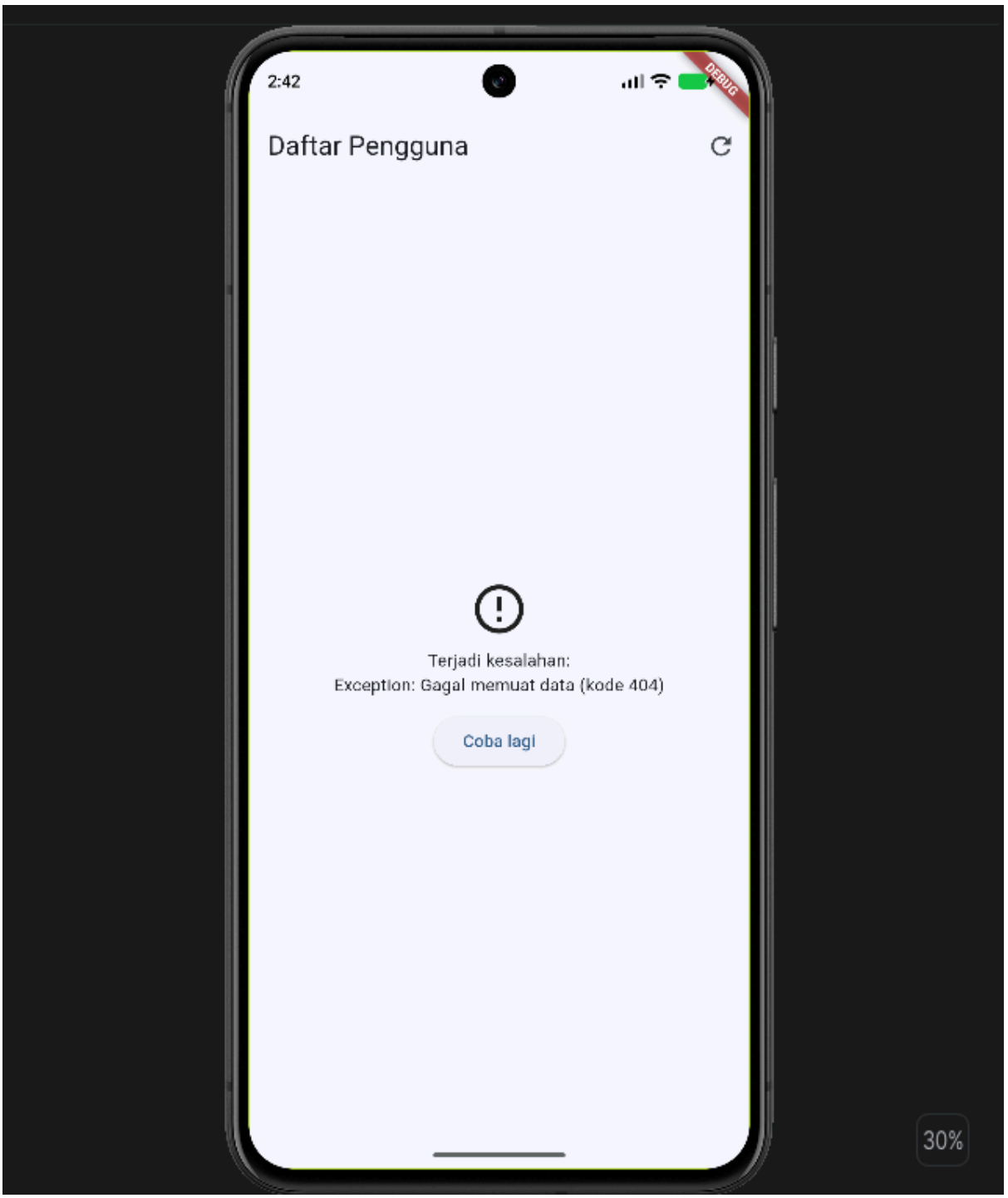
```

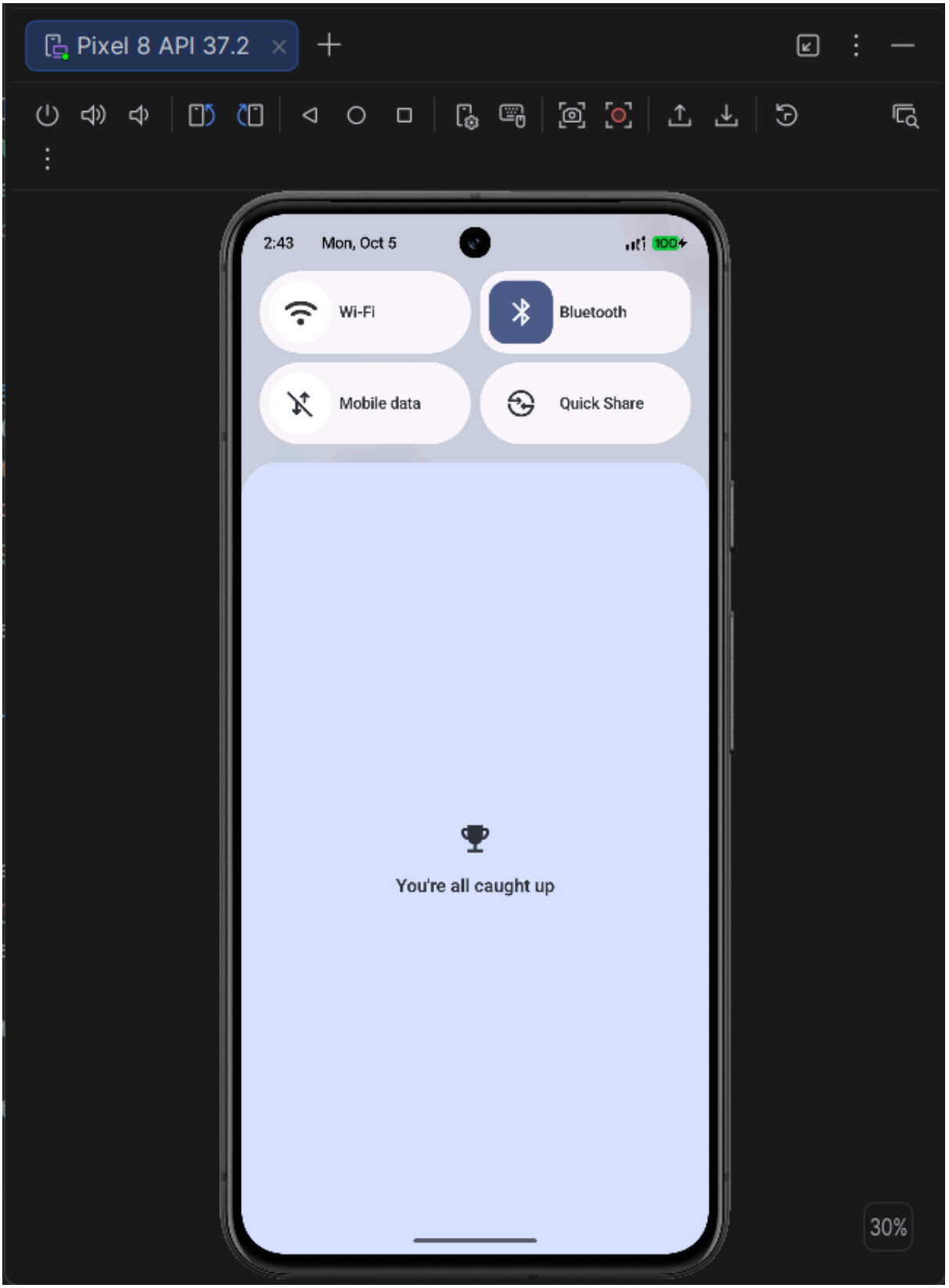

Pixel 8 API 37.2
2:38
DEBUG
Daftar Pengguna
L
Leanne Graham
Sincere@april.biz
E
Ervin Howell
Shanna@melissa.tv
C
Clementine Bauch
Nathan@yesenia.net
P
Patricia Lebsack
Julianne.OConner@kory.org
C
Chelsey Dietrich
Lucio_Hettinger@annie.ca
M
Mrs. Dennis Schulist
Karley_Dach@jasper.info
K
Kurtis Weissnat
Telly.Hoeger@billy.biz
N
Nicholas Runolfsdottir V
Sherwood@rosamond.me
G
Glenna Reichert
Chaim_McDermott@dana.io
C
Clementina DuBuque
Rey.Padberg@karina.biz
30%

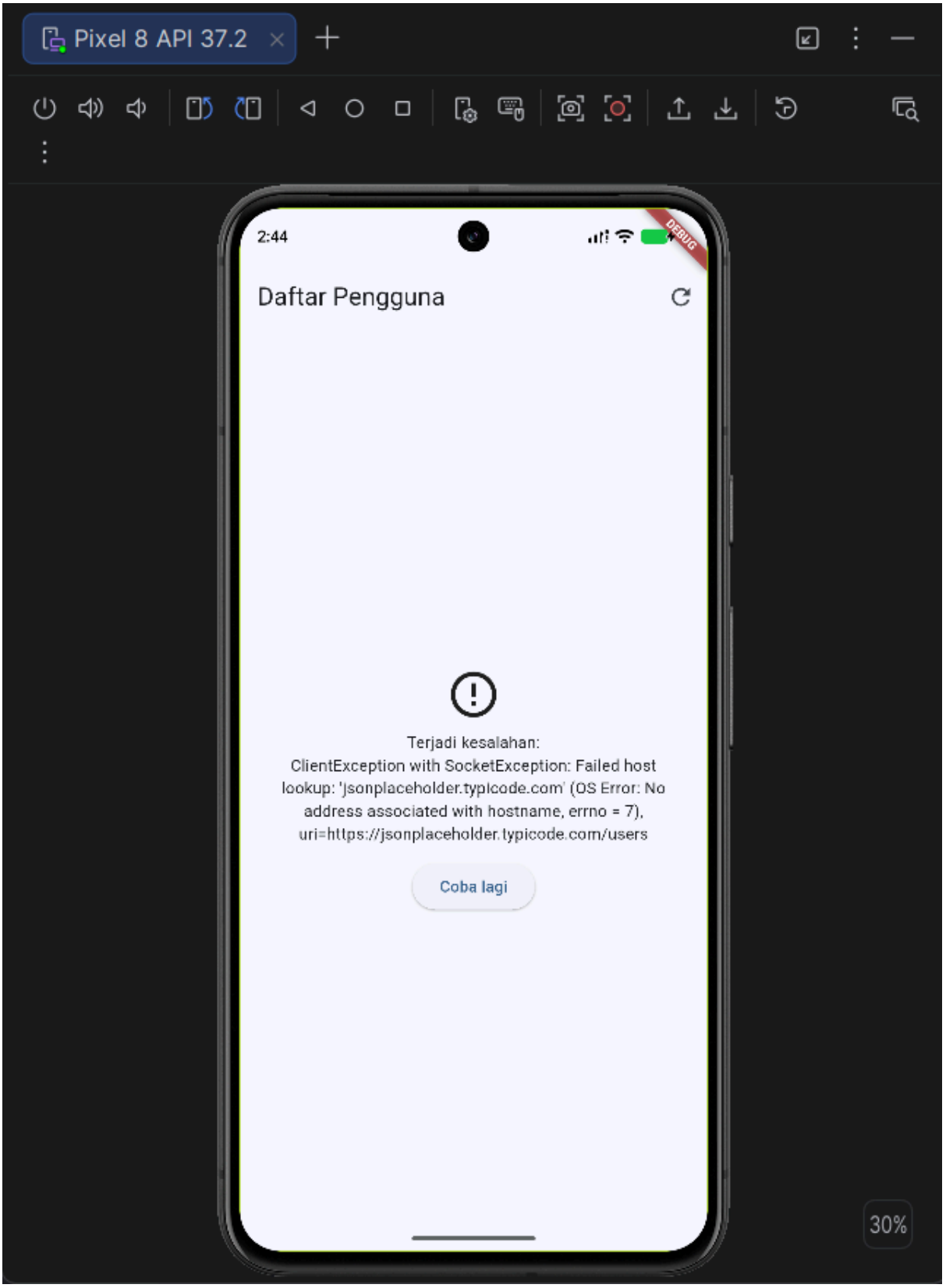

Pixel 8 API 37.2
2:39
DEBUG
Leanne Graham
Sincere@april.biz
1-770-736-8031 x56442
hildegard.org
30%

3.5 Percobaan Penanganan Galat :

```dart
Future<List<Pengguna>> ambilPengguna() async {
  final uri = Uri.parse('https://jsonplaceholder.typicode.com/userz');
  final response = await http.get(uri).timeout(const Duration(seconds: 10));
  if (response.statusCode != 200) {
    throw Exception('Gagal memuat data (kode ${response.statusCode})');
  }
  final List<dynamic> data = jsonDecode(response.body);
  return data
      .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
      .toList();
}
```




Pixel 8 API 37.2
2:43
Mon, Oct 5
Wi-Fi
Bluetooth
Mobile data
Quick Share
You're all caught up
30%


Pixel 8 API 37.2
2:44
DEBUG
Daftar Pengguna
Terjadi kesalahan:
ClientException with SocketException: Failed host lookup: 'jsonplaceholder.typicode.com' (OS Error: No address associated with hostname, errno = 7), uri=https://jsonplaceholder.typicode.com/users
Coba lagi
30%

4. Latihan Mandiri

1. Tambahkan kolom username dan kota (data bersarang: json['address']['city']) pada model dan tampilkan di halaman detail.
2. Bungkus ListView dengan RefreshIndicator agar dapat ditarik ke bawah untuk memuat ulang.
3. Bila daftar kosong, tampilkan teks "Tidak ada data".
4. Tambahkan Text di AppBar yang menampilkan jumlah pengguna setelah data berhasil dimuat.

main.dart

```dart
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

class Pengguna {
  final int id;
  final String name;
  final String username;
  final String email;
  final String phone;
  final String website;
  final String city;

  const Pengguna({
    required this.id,
    required this.name,
    required this.username,
    required this.email,
    required this.phone,
    required this.website,
    required this.city,
  });

  factory Pengguna.fromJson(Map<String, dynamic> json) {
    return Pengguna(
      id: json['id'] as int,
      name: json['name'] as String,
      username: json['username'] as String,
      email: json['email'] as String,
      phone: json['phone'] as String,
      website: json['website'] as String,
      city: json['address']['city'] as String,
    );
  }
}

Future<List<Pengguna>> ambilPengguna() async {
  final uri = Uri.parse(
```

```
    'https://jsonplaceholder.typicode.com/users',
  );

  final response = await http
      .get(uri)
      .timeout(const Duration(seconds: 10));

  if (response.statusCode != 200) {
    throw Exception(
      'Gagal memuat data (kode ${response.statusCode})',
    );
  }

  final List<dynamic> data = jsonDecode(response.body);

  return data
      .map(
        (e) => Pengguna.fromJson(
      e as Map<String, dynamic>,
    ),
  )
      .toList();
}

class DetailPenggunaPage extends StatelessWidget {
  final Pengguna pengguna;

  const DetailPenggunaPage({
```

```
    super.key,                                                    25
    required this.pengguna,
  });

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(pengguna.name),
      ), AppBar
      body: ListView(
        children: [
          ListTile(
            leading: const Icon(Icons.person),
            title: Text(pengguna.username),
          ), ListTile
          ListTile(
            leading: const Icon(Icons.email),
            title: Text(pengguna.email),
          ), ListTile
          ListTile(
            leading: const Icon(Icons.phone),
            title: Text(pengguna.phone),
```

```
            ), ListTile
            ListTile(
              leading: const Icon(Icons.language),
              title: Text(pengguna.website),
            ), ListTile
            ListTile(
              leading: const Icon(Icons.location_city),
              title: Text(pengguna.city),
            ), ListTile
          ],
        ), ListView
      ); Scaffold
    }
  }

  class PenggunaPage extends StatefulWidget {
    const PenggunaPage({super.key});

    @override
    State<PenggunaPage> createState() => _PenggunaPageState();
  }

  class _PenggunaPageState extends State<PenggunaPage> {
    late Future<List<Pengguna>> _future;

    @override
    void initState() {
      super.initState();
      _future = ambilPengguna();
    }

    void _muatUlang() {
      setState(() {
        _future = ambilPengguna();
      });
    }

    @override
    Widget build(BuildContext context) {
      return Scaffold(
        appBar: AppBar(
          title: FutureBuilder<List<Pengguna>>(
            future: _future,
            builder: (context, snapshot) {
              if (snapshot.hasData) {
                return Text(
```

```
136                     'Daftar Pengguna (${snapshot.data!.length})',
137                   ); Text
138                 }
139
140                 return const Text('Daftar Pengguna');
141               },
142             ), FutureBuilder
143             actions: [
144               IconButton(
145                 icon: const Icon(Icons.refresh),
146                 onPressed: _muatUlang,
147               ), IconButton
148             ],
149           ), AppBar
150           body: FutureBuilder<List<Pengguna>>(
151             future: _future,
152             builder: (context, snapshot) {
153               if (snapshot.connectionState ==
154                   ConnectionState.waiting) {
155                 return const Center(
156                   child: CircularProgressIndicator(),
157                 ); Center
158               }
159
160               if (snapshot.hasError) {
161                 return Center(
162                   child: Padding(
163                     padding: const EdgeInsets.all(24),
164                     child: Column(
165                       mainAxisSize: MainAxisSize.min,
```

```
166                       children: [
167                         const Icon(
168                           Icons.error_outline,
169                           size: 48,
170                         ), Icon
171                         const SizedBox(height: 8),
172                         Text(
173                           'Terjadi kesalahan:\n'
174                               '${snapshot.error}',
175                           textAlign: TextAlign.center,
176                         ), Text
177                         const SizedBox(height: 12),
178                         ElevatedButton(
179                           onPressed: _muatUlang,
180                           child: const Text('Coba lagi'),
181                         ), ElevatedButton
182                       ],
183                     ), Column
184                   ), Padding
```

```
185                 ); Center                                                    25 ^
186               }
187
188               final data = snapshot.data!;
189
190               if (data.isEmpty) {
191                 return const Center(
192                   child: Text('Tidak ada data'),
193                 ); Center
194               }
195
196               return RefreshIndicator(
197                 onRefresh: () async {
198                   _muatUlang();
199                   await _future;
200                 },
201                 child: ListView.builder(
202                   itemCount: data.length,
203                   itemBuilder: (context, i) {
204                     final p = data[i];
205
206                     return ListTile(
207                       leading: CircleAvatar(
208                         child: Text(p.name[0]),
209                       ), CircleAvatar
210                       title: Text(p.name),
211                       subtitle: Text(p.email),
212                       trailing: const Icon(
213                         Icons.chevron_right,
214                       ), Icon
```

```
215                       onTap: () {                                            25 ^
216                         Navigator.push(
217                           context,
218                           MaterialPageRoute(
219                             builder: (_) =>
220                                 DetailPenggunaPage(
221                                   pengguna: p,
222                                 ), DetailPenggunaPage
223                           ), MaterialPageRoute
224                         );
225                       },
226                     ); ListTile
227                   },
228                 ), ListView.builder
229               ); RefreshIndicator
230             },
```

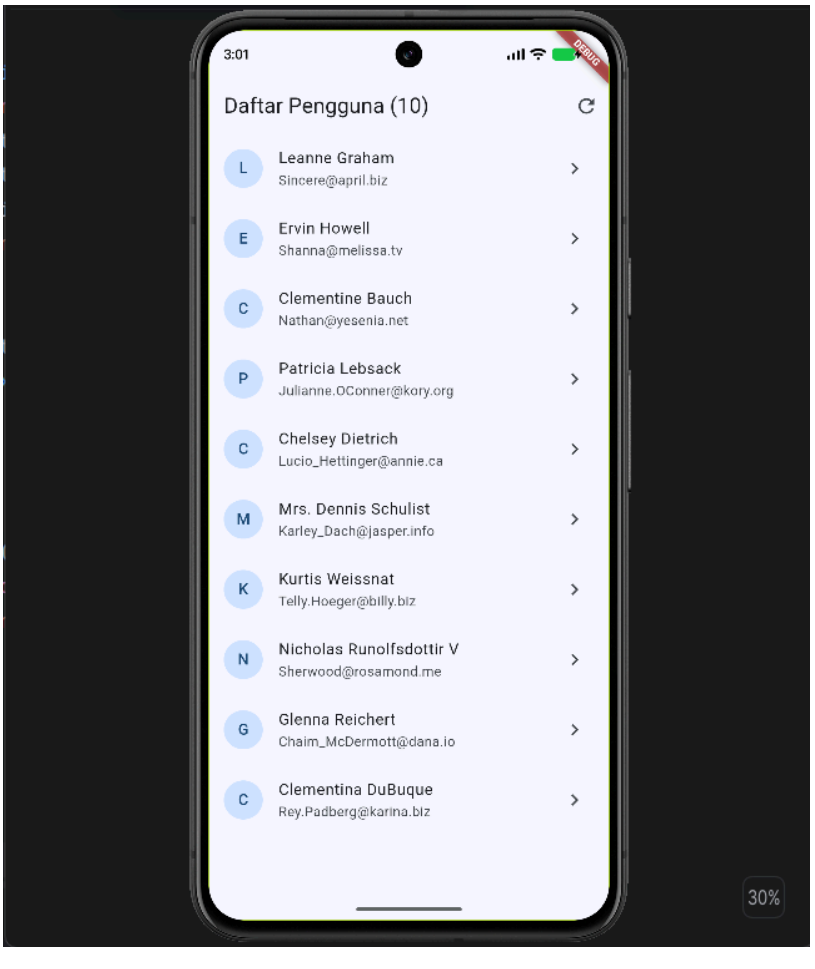
```
231           ), FutureBuilder
232         ); Scaffold
233       }
234     }
235
236     void main() => runApp(const MyApp());
237
238     class MyApp extends StatelessWidget {
239       const MyApp({super.key});
240
241       @override
242       Widget build(BuildContext context) {
243         return MaterialApp(
244           title: 'Praktikum 4',
245           theme: ThemeData(
246             colorSchemeSeed: Colors.blue,
247             useMaterial3: true,
248           ), ThemeData
249           home: const PenggunaPage(),
250         ); MaterialApp
251       }
252     }
253
254     Future<String> ambilSalam() async {
255       await Future.delayed(
256         const Duration(seconds: 2),
257       ); Future.delayed
258
259       return 'Halo dari masa depan!';
```

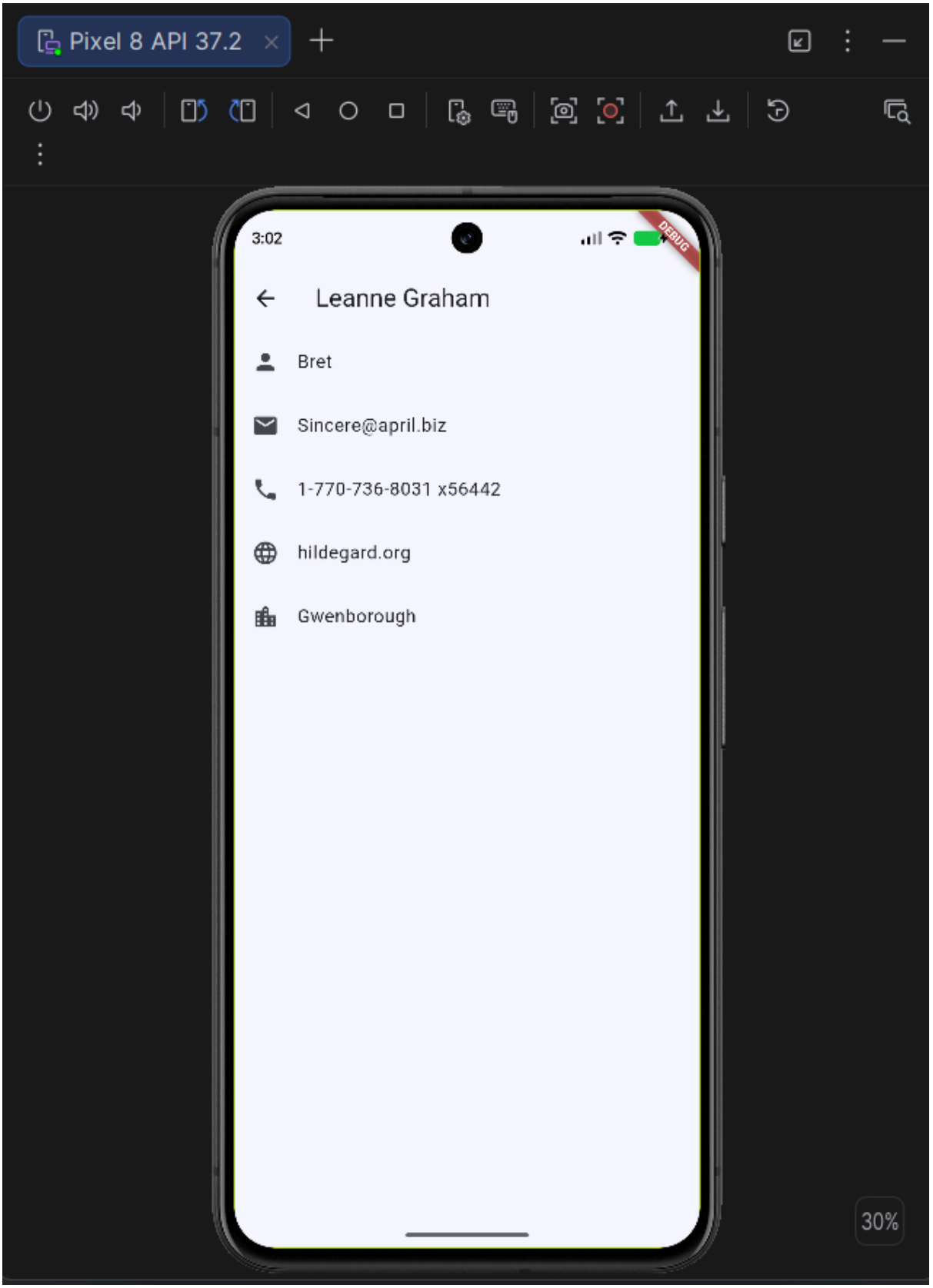
```
260     }                                                    25
261
262     class SalamPage extends StatefulWidget {
263       const SalamPage({super.key});
264
265       @override
266       State<SalamPage> createState() => _SalamPageState();
267     }
268
269     class _SalamPageState extends State<SalamPage> {
270       late Future<String> _future;
271
272       @override
273       void initState() {
274         super.initState();
275         _future = ambilSalam();
276       }
277
278       @override
```

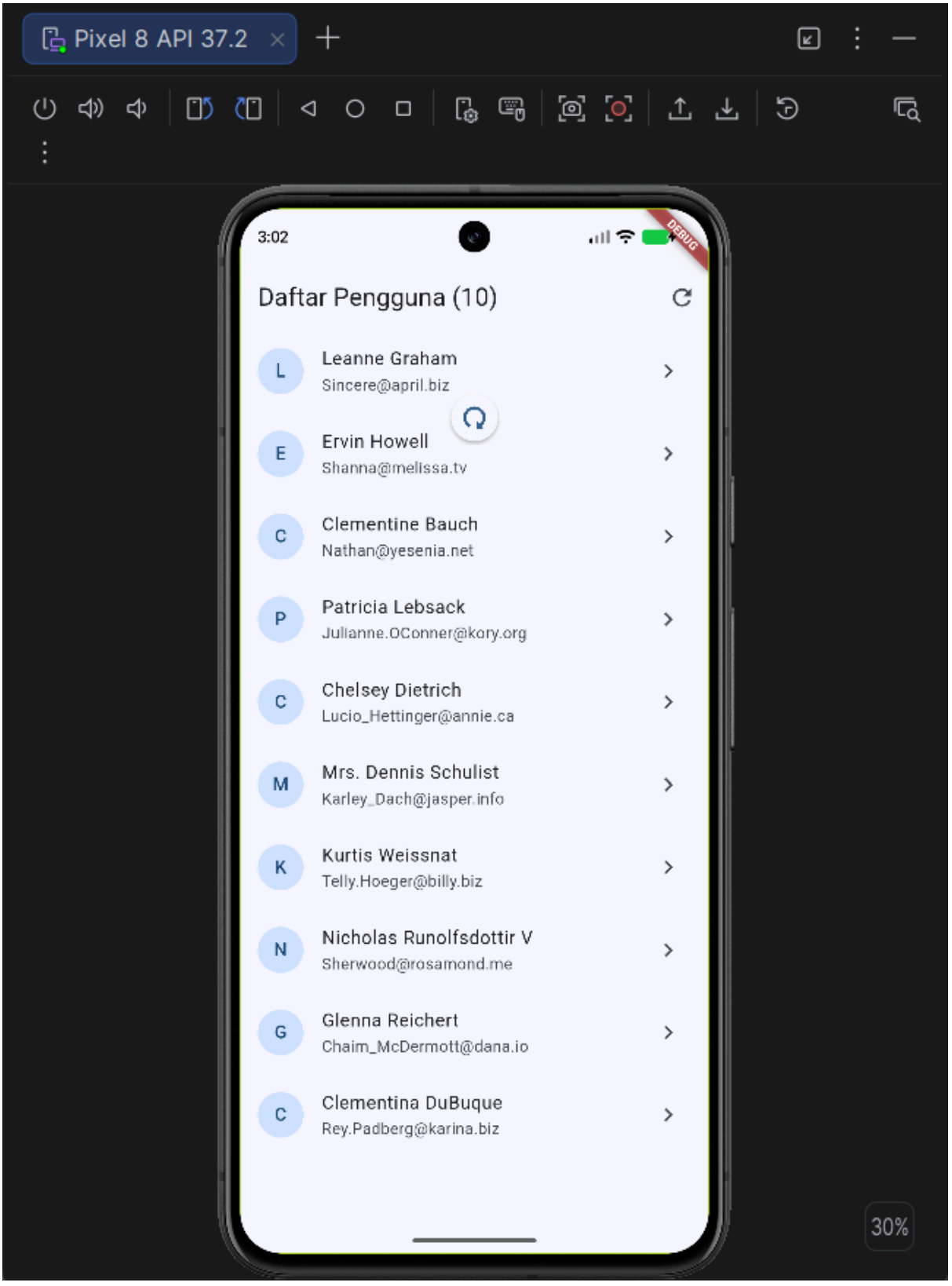
```dart
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Demo Future'),
    ), AppBar
    body: Center(
      child: FutureBuilder<String>(
        future: _future,
        builder: (context, snapshot) {
          if (snapshot.connectionState ==
              ConnectionState.waiting) {
            return const CircularProgressIndicator();
          }

          if (snapshot.hasError) {
            return Text(
              'Error: ${snapshot.error}',
            ); Text
          }

          return Text(
            snapshot.data!,
            style: const TextStyle(fontSize: 24),
          ); Text
        },
      ), FutureBuilder
    ), Center
  ); Scaffold
}
}
```




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Sincere@april.biz
1-770-736-8031 x56442
hildegard.org
Gwenborough
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Pixel 8 API 37.2
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Chelsey Dietrich
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M
Mrs. Dennis Schulist
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5. Tugas

Buat aplikasi Daftar Postingan menggunakan endpoint /posts dan /posts/{id}/comments dengan ketentuan:

- Halaman utama menampilkan daftar postingan (judul dan potongan isi) dengan FutureBuilder.
- Mengetuk postingan membuka halaman detail yang menampilkan isi lengkap dan daftar komentar yang diambil dari endpoint terpisah (Future kedua).
- Model Post dan Komentar dengan fromJson; kode pengambil data dipisah dari kode UI.
- Menangani status loading dan galat pada kedua halaman, termasuk tombol coba lagi.
- Kumpulkan: tangkapan layar (daftar, detail, dan tampilan galat) serta berkas main.dart (atau tautan repositori).

main.dart

```dart
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

class Post {
  final int id;
  final String title;
  final String body;

  const Post({
    required this.id,
    required this.title,
    required this.body,
  });

  factory Post.fromJson(Map<String, dynamic> json) {
    return Post(
      id: json['id'] as int,
      title: json['title'] as String,
      body: json['body'] as String,
    );
  }
}

class Komentar {
  final int id;
  final String name;
  final String email;
  final String body;
```

```dart
  const Komentar({
    required this.id,
    required this.name,
    required this.email,
    required this.body,
  });

  factory Komentar.fromJson(Map<String, dynamic> json) {
    return Komentar(
      id: json['id'] as int,
      name: json['name'] as String,
      email: json['email'] as String,
      body: json['body'] as String,
    );
  }
}

class ApiService {
  static const String baseUrl =
      'https://jsonplaceholder.typicode.com';

  Future<List<Post>> ambilPost() async {
    final uri = Uri.parse('$baseUrl/posts');
    final response = await http
        .get(uri)
        .timeout(const Duration(seconds: 10));

    if (response.statusCode != 200) {
      throw Exception(
        'Gagal memuat postingan (kode ${response.statusCode})',
      );
    }

    final List<dynamic> data = jsonDecode(response.body);

    return data
        .map(
          (e) => Post.fromJson(
        e as Map<String, dynamic>,
      ),
    )
        .toList();
  }

  Future<List<Komentar>> ambilKomentar(int postId) async {
    final uri = Uri.parse(
      '$baseUrl/posts/$postId/comments',
    );

```

```
    final response = await http
        .get(uri)
        .timeout(const Duration(seconds: 10));

    if (response.statusCode != 200) {
      throw Exception(
        'Gagal memuat komentar (kode ${response.statusCode})',
      );
    }

    final List<dynamic> data = jsonDecode(response.body);

    return data
        .map(
          (e) => Komentar.fromJson(
        e as Map<String, dynamic>,
      ),
    )
        .toList();
  }
}

class PostPage extends StatefulWidget {
  const PostPage({super.key});

  @override
  State<PostPage> createState() => _PostPageState();
}

class _PostPageState extends State<PostPage> {
```

```
  late Future<List<Post>> _future;

  @override
  void initState() {
    super.initState();
    _future = ApiService().ambilPost();
  }

  void _muatUlang() {
    setState(() {
      _future = ApiService().ambilPost();
    });
  }

  @override
  Widget build(BuildContext context) {
```

```
126     return Scaffold(
127       appBar: AppBar(
128         title: const Text('Daftar Postingan'),
129         actions: [
130           IconButton(
131             icon: const Icon(Icons.refresh),
132             onPressed: _muatUlang,
133           ), IconButton
134         ],
135       ), AppBar
136       body: FutureBuilder<List<Post>>(
137         future: _future,
138         builder: (context, snapshot) {
139           if (snapshot.connectionState ==
140               ConnectionState.waiting) {
141             return const Center(
142               child: CircularProgressIndicator(),
143             ); Center
144           }
145
146           if (snapshot.hasError) {
147             return Center(
148               child: Padding(
149                 padding: const EdgeInsets.all(24),
150                 child: Column(
151                   mainAxisSize: MainAxisSize.min,
152                   children: [
153                     const Icon(
154                       Icons.error_outline,
```

```
                    size: 48,
                  ), Icon
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n${snapshot.error}',
                    textAlign: TextAlign.center,
                  ), Text
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ), ElevatedButton
                ],
              ), Column
            ), Padding
          ); Center
        }

        final data = snapshot.data!;

        return ListView.builder(
          itemCount: data.length,
          itemBuilder: (context, i) {
            final post = data[i];

            return ListTile(
              title: Text(post.title),
              subtitle: Text(
                post.body,
                maxLines: 2,
                overflow: TextOverflow.ellipsis,
              ), Text
              trailing: const Icon(
                Icons.chevron_right,
              ), Icon
              onTap: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(
                    builder: (_) => DetailPostPage(
                      post: post,
                    ), DetailPostPage
                  ), MaterialPageRoute
                );
              },
            ); ListTile
          },
        ); ListView.builder
      },
    ), FutureBuilder
  ); Scaffold
}
```

```dart
}

class DetailPostPage extends StatefulWidget {
  final Post post;

  const DetailPostPage({
    super.key,
    required this.post,
  });

  @override
  State<DetailPostPage> createState() =>
      _DetailPostPageState();
}

class _DetailPostPageState extends State<DetailPostPage> {
  late Future<List<Komentar>> _future;

  @override
  void initState() {
    super.initState();
    _future = ApiService().ambilKomentar(
      widget.post.id,
    );
  }

  void _muatUlang() {
    setState(() {
      _future = ApiService().ambilKomentar(
        widget.post.id,
      );
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Detail Postingan'),
        actions: [
          IconButton(
            icon: const Icon(Icons.refresh),
            onPressed: _muatUlang,
          ), IconButton
        ],
      ), AppBar
```

```
253        body: ListView(
254          padding: const EdgeInsets.all(16),
255          children: [
256            Text(
257              widget.post.title,
258              style: const TextStyle(
259                fontSize: 22,
260                fontWeight: FontWeight.bold,
261              ), TextStyle
262            ), Text
263            const SizedBox(height: 12),
264            Text(widget.post.body),
265            const SizedBox(height: 24),
266            const Text(
267              'Komentar',
268              style: TextStyle(
269                fontSize: 20,
270                fontWeight: FontWeight.bold,
271              ), TextStyle
272            ), Text
273            const SizedBox(height: 8),
274            FutureBuilder<List<Komentar>>(
275              future: _future,
276              builder: (context, snapshot) {
277                if (snapshot.connectionState ==
278                    ConnectionState.waiting) {
279                  return const Center(
280                    child: Padding(
281                      padding: EdgeInsets.all(24),
282                      child: CircularProgressIndicator(),
283                    ), Padding
284                  ); Center
285                }
286
287                if (snapshot.hasError) {
288                  return Column(
289                    children: [
290                      const Icon(
291                        Icons.error_outline,
292                        size: 48,
293                      ), Icon
294                      const SizedBox(height: 8),
295                      Text(
296                        'Terjadi kesalahan:\n${snapshot.error}',
297                        textAlign: TextAlign.center,
298                      ), Text
299                      const SizedBox(height: 12),
300                      ElevatedButton(
```

27

```
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ), ElevatedButton
                ],
              ); Column
            }

            final komentar = snapshot.data!;

            return ListView.builder(
              shrinkWrap: true,
              physics:
              const NeverScrollableScrollPhysics(),
              itemCount: komentar.length,
              itemBuilder: (context, i) {
                final k = komentar[i];

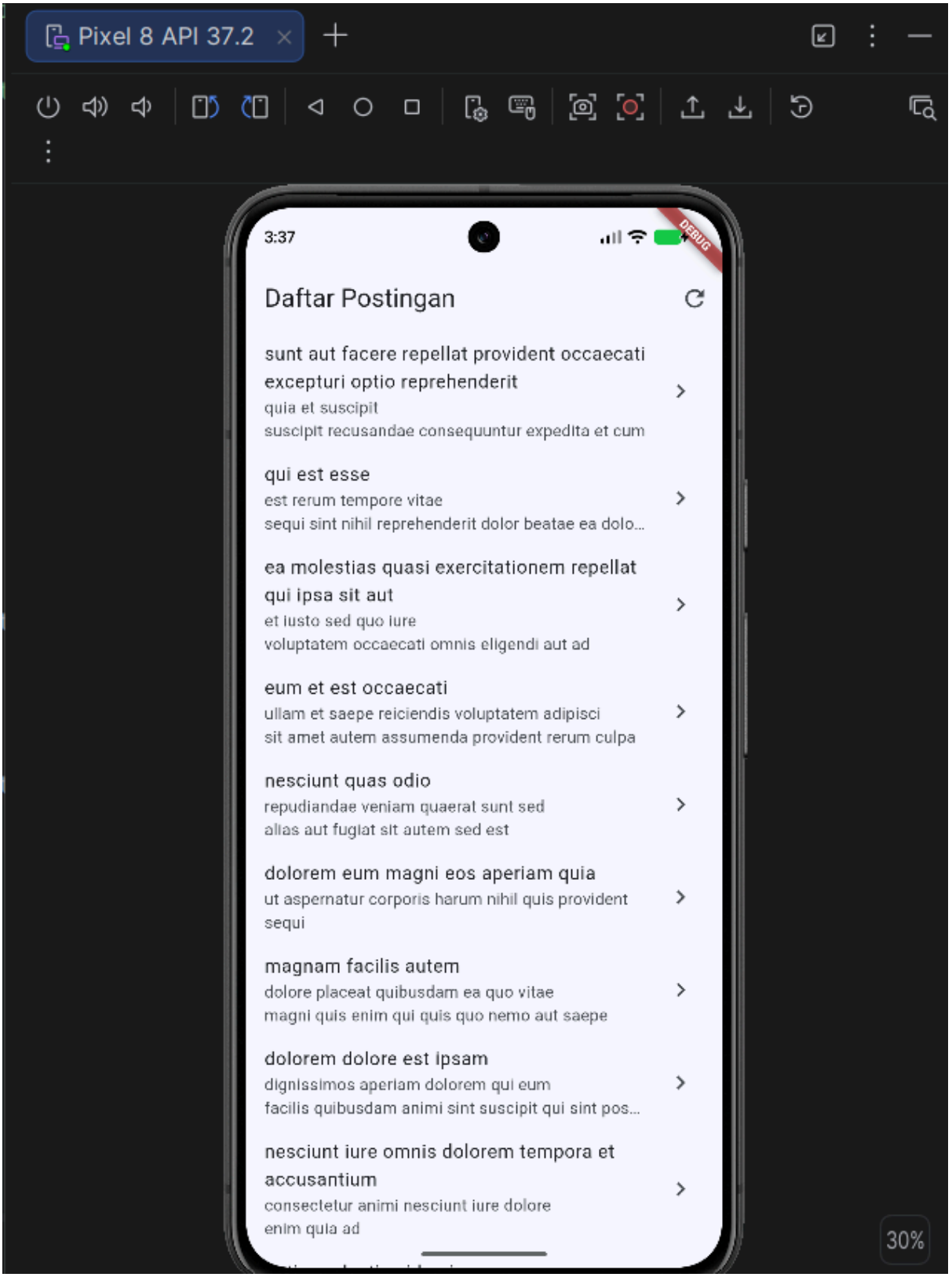
                return ListTile(
                  contentPadding: EdgeInsets.zero,
                  leading: const CircleAvatar(
                    child: Icon(Icons.person),
                  ), CircleAvatar
                  title: Text(k.name),
                  subtitle: Text(
                    '${k.email}\n\n${k.body}',
                  ), Text
                ); ListTile
              },
```

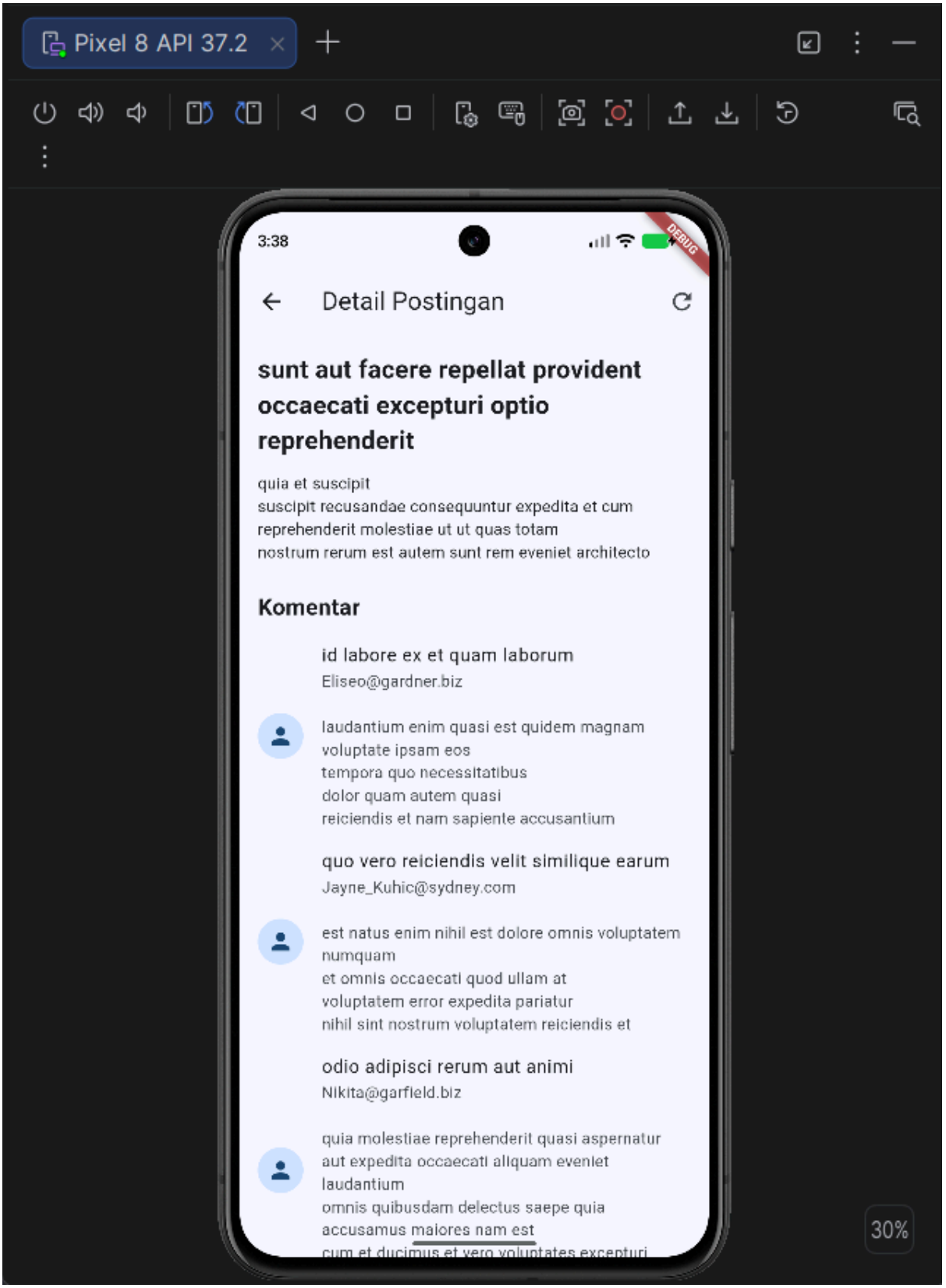
```
            ); ListView.builder                                    27
          },
        ), FutureBuilder
      ],
    ), ListView
  ); Scaffold
 }
}

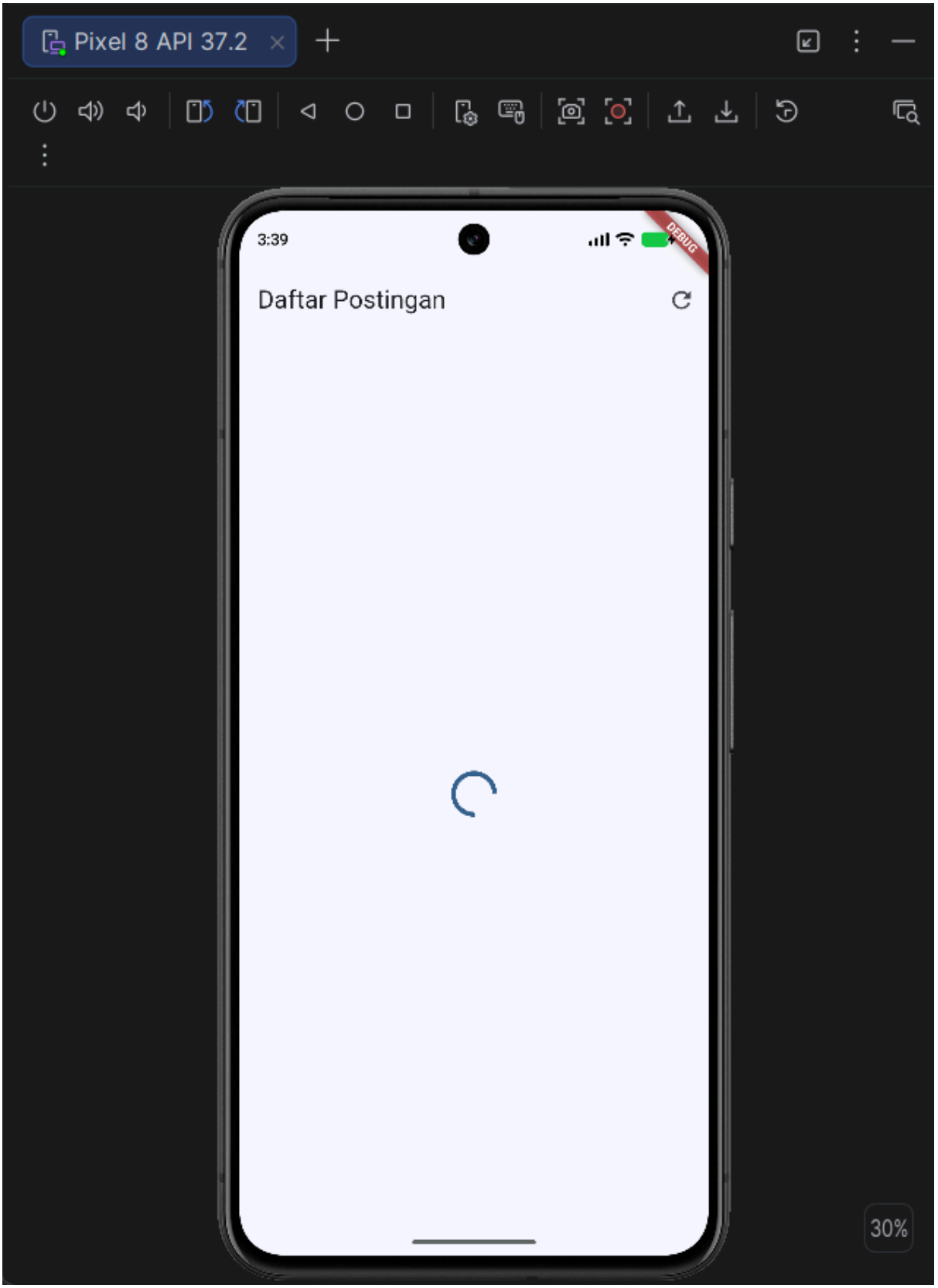
void main() {
 runApp(const MyApp());
}

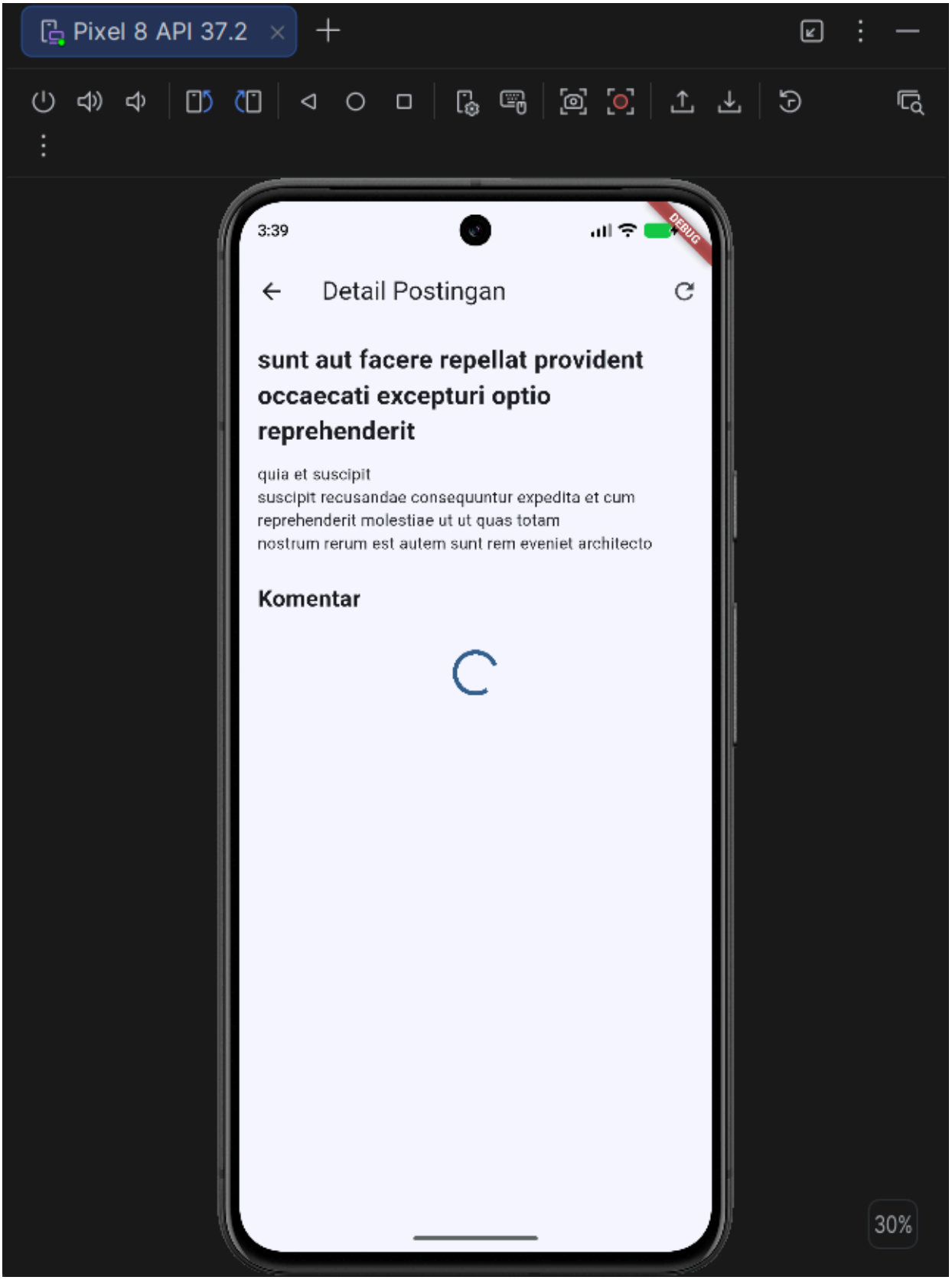
class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   title: 'Praktikum 4',
   theme: ThemeData(
    colorSchemeSeed: Colors.blue,
    useMaterial3: true,
   ), ThemeData
   home: const PostPage(),
  ); MaterialApp
 }
}
```


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6. Pertanyaan Refleksi

1. Apa yang terjadi bila Future dibuat di dalam build() dan bukan di initState()? Mengapa?

Jika Future dibuat di dalam build(), Future dapat dibuat ulang setiap kali widget mengalami rebuild. Akibatnya proses pengambilan data dapat berjalan berulang-ulang.

2. Apa perbedaan snapshot.hasError dengan memeriksa response.statusCode? Mengapa keduanya diperlukan?

response.statusCode digunakan untuk memeriksa apakah server memberikan respons HTTP yang berhasil, misalnya 200, atau memberikan kode seperti 404 dan 500. Sedangkan snapshot.hasError digunakan untuk mengetahui apakah Future menghasilkan error, termasuk error yang terjadi ketika proses pengambilan data gagal.

3. Mengapa data JSON sebaiknya diubah menjadi class model, bukan dipakai langsung sebagai Map?

Data JSON sebaiknya diubah menjadi class model agar struktur data lebih jelas dan mudah digunakan di dalam aplikasi. Dengan Pengguna, misalnya, kita bisa langsung menggunakan pengguna.name, pengguna.email, dan pengguna.phone daripada mengakses key pada Map berulang kali.

4. Mengapa UI wajib menyediakan status loading dan error, bukan hanya status data?

Karena pengambilan data dari internet membutuhkan waktu dan dapat mengalami kegagalan. Status loading memberitahu pengguna bahwa data masih sedang dimuat, sedangkan status error memberitahu bahwa pengambilan data gagal.

7. Catatan Pribadi

Future : Objek yang berisi hasil asynchronous yang akan tersedia nanti.

async : Menandakan fungsi melakukan operasi asynchronous.

await : Menunggu hasil proses asynchronous tanpa memblokir UI.

REST API : Layanan yang memungkinkan aplikasi bertukar data melalui HTTP.

HTTP GET : Request untuk mengambil data dari server.

JSON : Format data yang digunakan API untuk mengirim atau menerima informasi.

jsonDecode() : Mengubah teks JSON menjadi Map atau List Dart.

fromJson : Mengubah data JSON berbentuk Map menjadi object model.

http.get() : Mengirim request GET ke URL API.

FutureBuilder : Widget untuk membangun UI berdasarkan keadaan sebuah Future.

snapshot : Menyimpan status, data, dan error dari proses Future.

ConnectionState.waiting : Menandakan data masih dalam proses dimuat.

snapshot.hasError : Menandakan Future menghasilkan error.

snapshot.data : Berisi data hasil Future yang berhasil diperoleh.